Robust Learning Under Label Noise for Real-World Classification

Evaluation & Reproducibility Report | Experiment ID: 5b1ee218-efcb-4a9a-84d9-49349802b86e | Generated: 2026-10-26 23:25:18

1. Dataset & Experimental Setup

Target Variable:	diagnosis	**Classes:**	2 (Benign, Malignant)
Total Rows:	569	**Preprocessed Features:**	30
Train / Test Split:	399 (70%) / 85 (15%)	**Noise Levels Evaluated:**	0%, 20%
Random Seed:	[42]	**Noise Strategy:**	Symmetric

2. Model Evaluation Across Noise Levels

Evaluated strictly on the clean holdout test set to measure genuine generalization ability under synthetic training noise.

Model	Type	Noise	Accuracy	Precision	Recall	F1-Score	Time (s)
Baseline Logistic Regression	Baseline	0%	1.0000	1.0000	1.0000	**1.0000**	0.00
Robust Sample-Weighted (Logistic Regression)	Robust	0%	1.0000	1.0000	1.0000	**1.0000**	0.01
Baseline Logistic Regression	Baseline	20%	0.9294	0.9319	0.9294	**0.9283**	0.01
Robust Sample-Weighted (Logistic Regression)	Robust	20%	0.9294	0.9319	0.9294	**0.9283**	0.01

3. Confident Learning Label Issue Detection

Cross-validation out-of-fold probability estimation identified **72** suspicious training samples (18.05% of training split) exceeding the issue threshold of 0.50. These samples exhibited substantial margin disagreement between the assigned label and the learned distribution.

Row ID	Observed Label	Predicted Class	Confidence	Observed Prob	Issue Score	Status
213	Malignant	Benign	0.99	0.01	**0.98**	Potential Issue
3	Benign	Malignant	0.98	0.02	**0.97**	Potential Issue
352	Benign	Malignant	0.97	0.03	**0.96**	Potential Issue
290	Benign	Malignant	0.95	0.05	**0.93**	Potential Issue
176	Malignant	Benign	0.93	0.07	**0.90**	Potential Issue
164	Benign	Malignant	0.92	0.08	**0.88**	Potential Issue
546	Malignant	Benign	0.92	0.08	**0.88**	Potential Issue
524	Malignant	Benign	0.91	0.09	**0.87**	Potential Issue

4. Empirical Findings & Discussion

• **Baseline Performance:** Under pristine labels (0% noise), Baseline Logistic Regression achieved the highest F1-score of 1.0000 (Accuracy: 1.0000).

• **Noise Robustness:** At 20% label noise, robust and conventional classifiers performed comparably (Robust Sample-Weighted (Logistic Regression): 0.9283 vs Baseline Logistic Regression: 0.9283).

• **Noise Degradation:** Baseline Logistic Regression experienced an F1 degradation of 0.0717 as noise increased from 0% to 20%.

5. Methodological Notes & Reproducibility

Reproducibility: Results are strictly deterministic given the configured random seed and split ratios. Label noise was applied solely to the training set; test labels remained unaltered to ensure valid evaluation.
Technical Note: Confident learning scores indicate statistical discrepancy between observed labels and feature density; they represent potential label noise, not mathematical proof of label error in uncontrolled environments.